Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Reply to responses on paper 1

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Sun, Sep 27, 2026 at 6:00 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>

Ben,

Thank you for the final version and for the detailed implementation of the amendments.

The lemma now reads on regular conditional distributions with the bound for almost every y, the proof names μ_y and ν_{y'} as such,
and the uniformity in x is what carries it (that is the right repair and it leaves the argument untouched).
The rate is "at least linearly" in all three places, and the added sentence that the lemma gives no lower bound,
so a system's defect may vanish faster than linearly or vanish identically, is worth more than the amendment I asked for.
Persistence is conditional in Section 2. The sampling floor is in Section 6 as a requirement to declare and calibrate rather than derive,
and you were right to leave my δ_min formula out: I gave it as a conditional heuristic and the paper is better without a formula it cannot
support. Λ_0 and Λ are separated, with the bare theory at a point as the $\Lambda_0 \to \infty$ end of the flow.
The time-slice sentence stops where it should. Definition 1(a) says "uniquely solvable." The modal language is consistent.

On Wightman: you went to the source (I had confirmed the bibliographic entry only).

On the invariant, you are right and the correction is the substantive result of this exchange. I posed the open problem as the absence
of a description-independent invariant. By the fact I myself cited, no such invariant can exist: with $\tilde{X}_t = (X_s)_{s\le t}$ one has $\sigma(\tilde{X}_s) = F_s$,
so the path-space description is Markov for every process and its defect vanishes identically; any quantity nonzero somewhere and invariant
under change of description would have to vanish there. So what I called a missing result was a problem stated in an unattainable form.
Your formulation (an invariant relative to a physically fixed class of admissible descriptions, with the class as the open problem) is more suitable,
and Section 7 now records it in the form in which it can be worked on.

One addition, which I think sharpens what remains. Neither of the two constraints the paper supplies acts in the direction where the impossibility bites.
The resolution bound restricts operational access in time; the non-invasiveness condition restricts the observation protocol.
Neither bounds history depth or state-space dimension, and a path-space state on a coarse time grid respects the Planck bound, is non-invasive,
and has zero defect. The class is unpinned precisely where history augmentation operates, and that is the sharpest statement
I can give of the problem your portfolio leaves open.

I think that the review cycle on this paper could be closed favourably, with essentially nothing outstanding.

Best,
Ezequiel

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[Quoted text hidden]